

Math 421

Friday, September 26

Announcements

- Homework 3 due **today**

Additional Consequences of Completeness

Thm (Density of $\mathbb{Q}$). If $a, b \in \mathbb{R}$ and $a < b$, then there is a rational $r \in \mathbb{Q}$ such that $a < r < b$. *in $\mathbb{R}$*
 $r \in (a, b)$

Corollary: there are infinitely many rational numbers in any open interval of $\mathbb{R}$.

Thm (Existence of Square Roots). If $x \geq 0$, then there exists a real number $\alpha \in \mathbb{R}$ satisfying $\alpha \geq 0$ and $\alpha^2 = x$ (i.e. $\sqrt{x}$ exists).

Recipe to Show $\sup S = M$

least upper bound

Step 1: Prove that M is an upper bound for S (i.e show $M \geq s$ for all $s \in S$) and therefore $\sup S$ exists.

Step 2: Prove that M is the least upper bound by using one of the following methods.

- Method 1: Assume there exists another upper bound, M_0 , prove that $M \leq M_0$.
- Method 2: Show that for all $\epsilon > 0$, there exists $s \in S$ such that $s > M - \epsilon$.
 $M - \epsilon < s \leq M$

$$A = \{2, 3\} \quad B = \{4, 9\}$$

Example 1: Proof Sketch Using Recipe

Theorem. Assume $A \subseteq [0, \infty)$ is nonempty and bounded above. If $B = \{a^2 : a \in A\}$, then $\sup B = (\sup A)^2$.

Proof Sketch. Step 1: $a \leq \sup A \Rightarrow a^2 \leq (\sup A)^2$ for all $a \in A$

So $(\sup A)^2$ is an upper bound for B . B is nonempty.

$\sup B$ exists.

Step 2: (Method 1) let M_0 be another upper bound for B .
 $a^2 \leq M_0 \Rightarrow a \leq \sqrt{M_0}$ so $\sqrt{M_0}$ is an upper bound for A .

$$\Rightarrow \sup A \leq \sqrt{M_0} \Rightarrow \underbrace{(\sup A)^2}_{\leq M_0}$$

$$\Rightarrow \sup B = (\sup A)^2$$

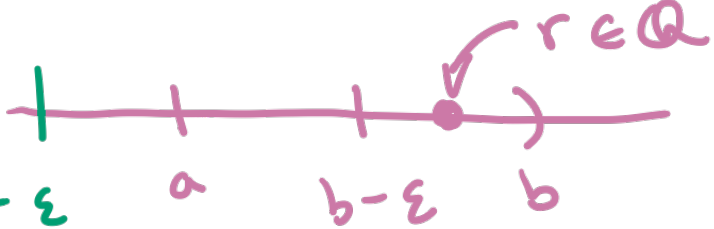
Method 2: $\forall \varepsilon > 0$ there exists $s \in A$ s.t.
 $b - \varepsilon < s$.

Example 2: Proof Sketch Using Recipe

Theorem. Let $A = [a, b)$. Then $\sup A = b$.

Proof Sketch. Step 1: $x \in A$, $a \leq x < b$ so b is an upper bound for A . A is nonempty. So $\sup A$ exists.

Step 2: (Method 2):



case 1: $b - \varepsilon < a$. Then $a \in A$ and $b - \varepsilon < a < b$.

case 2: $b - \varepsilon \geq a$. Then, by density, there exists $r \in \mathbb{Q}$ such that $a \leq b - \varepsilon < r < b$ so $r \in A$.

$\implies \sup A = b$.

Chapter 2: Sequences

Section 7: Limits of Sequences

Sequences

Definition. A **sequence** is a function that has a specified value for each integer $n \geq m$ (usually m is either 0 or 1).

Notation: $(s_n)_{n \in \mathbb{N}}$ or $(s_1, s_2, s_3, \dots)$

$(s_n) \rightarrow$ assumed to be $(s_n)_{n \in \mathbb{N}}$

Examples:

1. $(s_n)_{n \in \mathbb{N}}$ where $s_n = \frac{1}{n^2}$ $(s_1, s_2, s_3, \dots) = (1, \frac{1}{2^2}, \frac{1}{3^2}, \dots)$ set of values $\{1, \frac{1}{4}, \frac{1}{9}, \dots\}$
2. $(s_n)_{n=0}^{\infty}$ where $s_n = \sin \frac{n\pi}{4}$ $(0, \frac{\sqrt{2}}{2}, 1, \frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}, -1, -\frac{\sqrt{2}}{2}, 0, \dots)$
set of values: $\{0, \frac{\sqrt{2}}{2}, 1, -\frac{\sqrt{2}}{2}, -1\}$

Informally: ... the terms s_n approach s as n gets larger & larger.

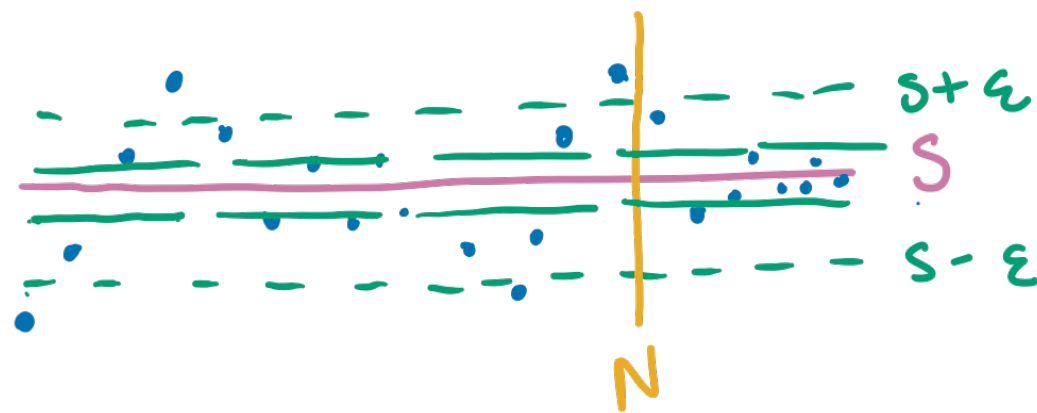
Convergence

Def: A sequence (s_n) of real numbers is said to **converge** to the real number s provided that:

$$\lim_{n \rightarrow \infty} s_n = s \quad \text{or} \quad (s_n) \rightarrow s$$

For each $\epsilon > 0$ there exists a number N such that whenever $n > N$ implies $|s_n - s| < \epsilon$.

$$s_n \in (s - \epsilon, s + \epsilon)$$



Activity

Give an example of the following:

1. Two different sequences that have the same *set of values*.

$$\left(\sin \left(\frac{n\pi}{4} \right) \right) \quad \left(\cos \left(\frac{n\pi}{4} \right) \right)$$

2. Two different sequences that converge to the same real number s .

$$\left(\frac{1}{n} \right) \quad \left(\frac{1}{n^2} \right).$$